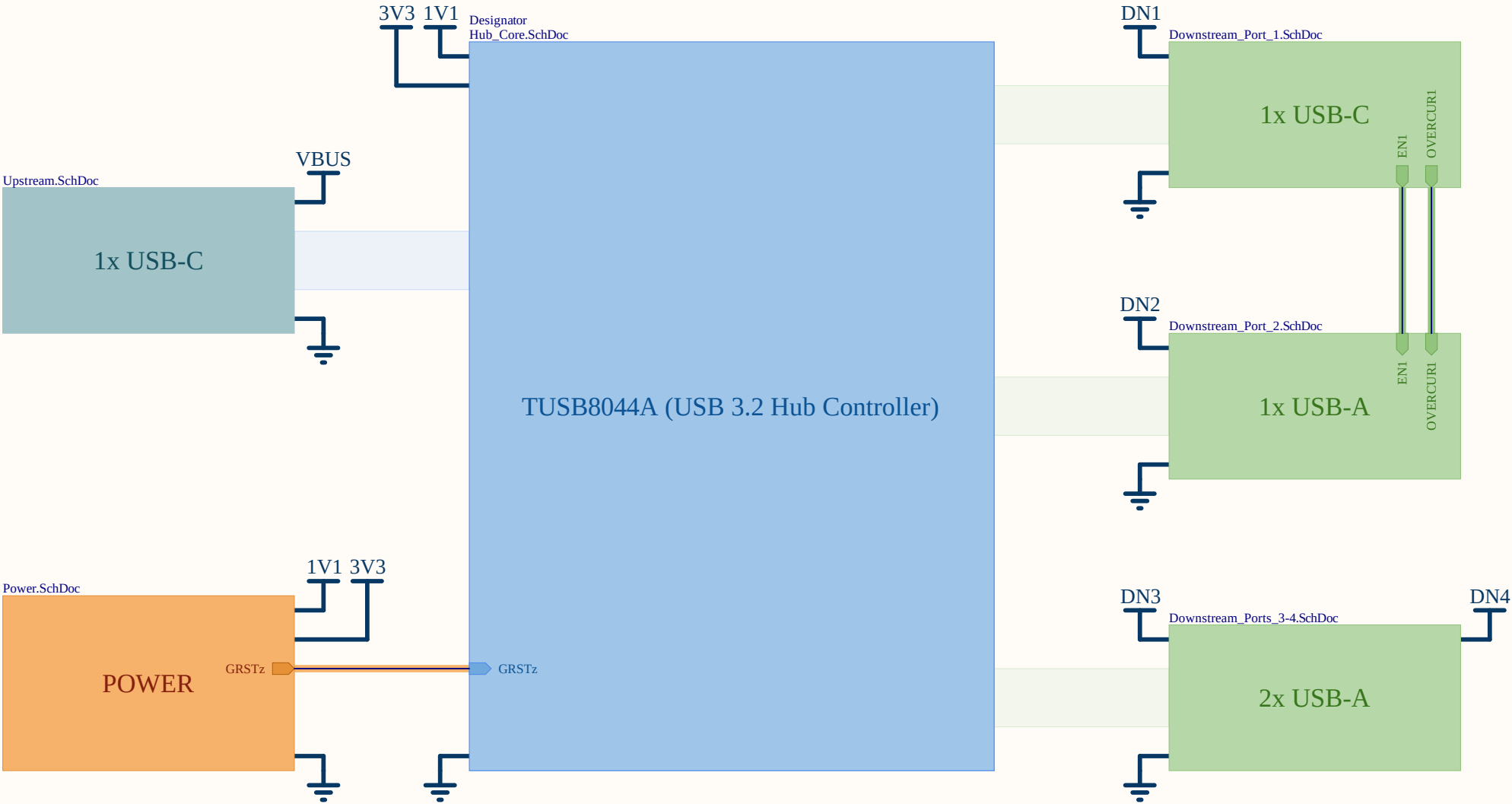


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USB 3.2 Gen 1 4-Port Hub



POWER BUDGET - USB 3.2 Gen1 Hub Bus-Powered, No USB PD		
INPUT		
USB-C Upstream (5V, 3A max)	15.00 W	3000 mA
HUB CONSUMPTION		
TUSB8044A VDD (1.1V x 680mA)	0.75 W	680 mA*
TUSB8044A VDD33 (3.3V x 100mA)	0.33 W	100 mA*
LDO Loss (2.2V x 680mA)	1.50 W	---
Buck Loss (5V->3.3V, eff~90%)	0.28 W	---
HD3SS3220 x2	0.01 W	2 mA
Misc (pull-ups, LED)	0.02 W	5 mA
Total from 5V	2.91 W	582 mA
* Current on respective rails		
AVAILABLE FOR DOWNSTREAM		
3000 - 582 = 2418 mA @ 5V	12.09 W	2418 mA
PORT CURRENT LIMITS (Per Channel)		
P1 USB-C (TPS2561#1) R_ILIM=37.4k		1.5 A max
P2 USB-A (TPS2561#1) R_ILIM=37.4k		1.5 A max
P3 USB-A (TPS2561#2) R_ILIM=56.0k		1.0 A max
P4 USB-A (TPS2561#2) R_ILIM=56.0k		1.0 A max
Worst case all ports:	5.0 A	[EXCEEDS BUDGET]
Typical 2 ports:	~1.0 A	[WITHIN BUDGET]
NOTES		
- Worst case exceeds 3A upstream. PC Host overcurrent protection will limit total draw.		
- BC 1.2 CDP: 1.5A/port while data active.		
- BC 1.2 DCP: 1.5A/port, upstream absent.		

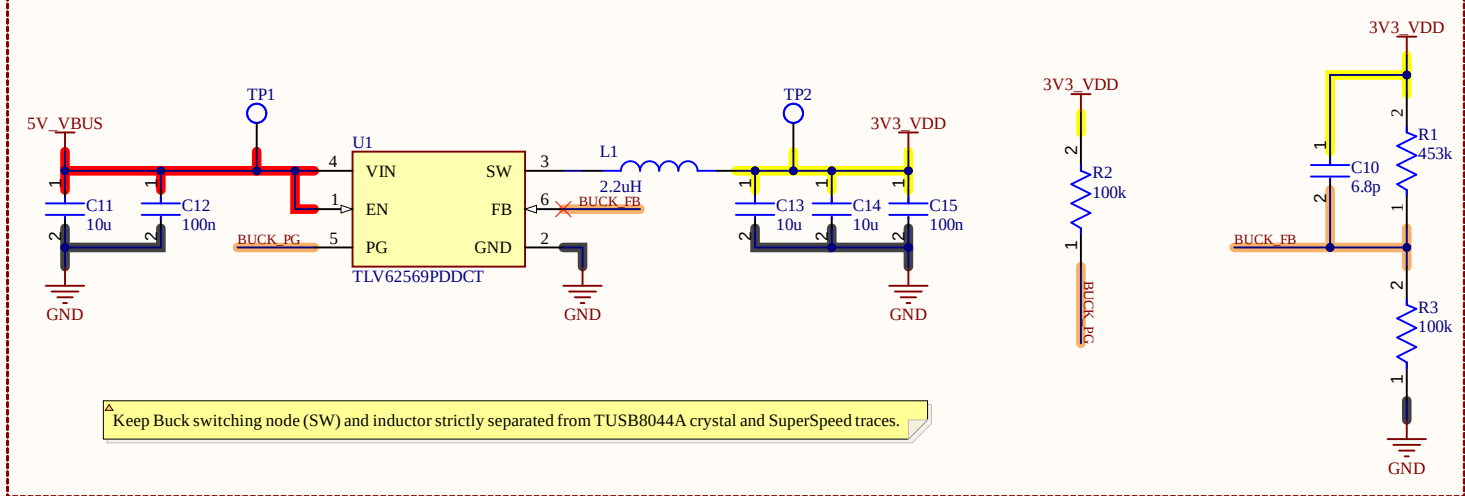
DESIGN NOTES
SUPERSPEED ROUTING OPTIMIZATION RX/TX polarity inversion (P/N swapping) has been implemented on specific downstream/upstream ports to optimize PCB routing and preserve Signal Integrity. This is fully compliant with the USB 3.2 Gen 1 specification, which mandates automatic polarity inversion detection at the receiver.

APPROVALS		DATE	PROJECT		Altium™	
ENG:	Alberto Marrone					
DSN:	Alberto Marrone		PROJECT REVISION:	DOCUMENT REVISION:	DESIGN ITEM:	
CHK:			Not in version control		Not in version control	
REFERENCE DOCUMENTS			TITLE			
BOM:			USB 3.2 Gen 1 4-Port Hub			
ASSY DWG:			SIZE	CAGE CODE	DWG NO.	REV
FAB DWG:			B			
PCB DWG:			SCALE:	FILE NAME		SHEET
				Top_Level.SchDoc		1 OF 7

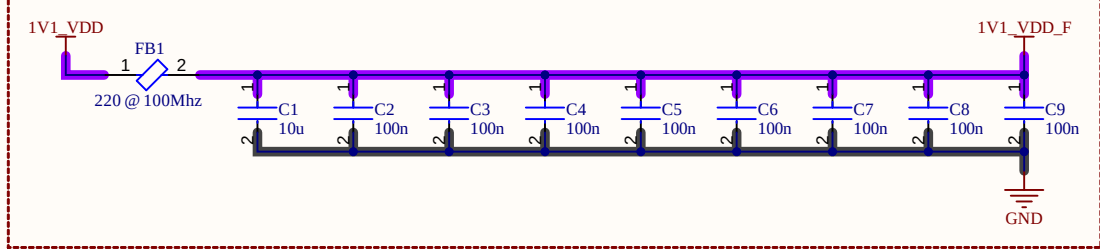
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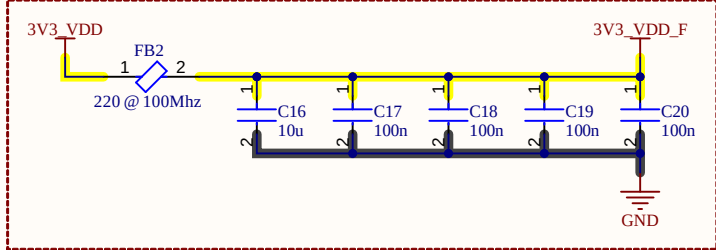
5V to 3.3V Buck Converter



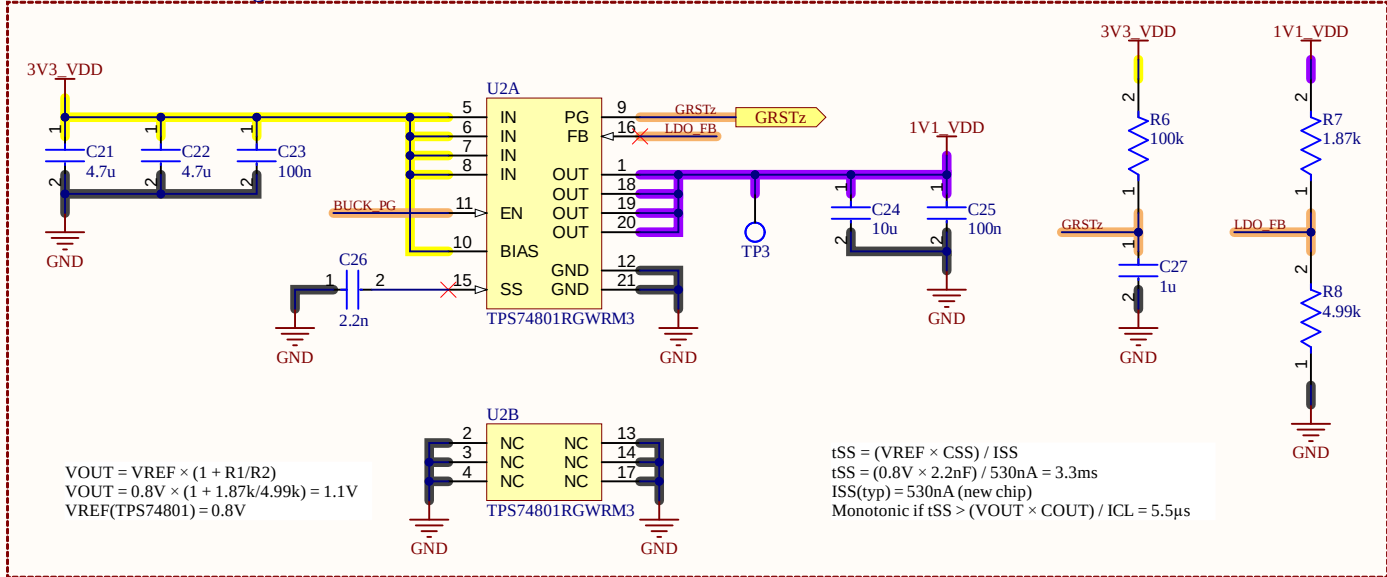
TUSB8044A Decoupling - VDD 1.1V



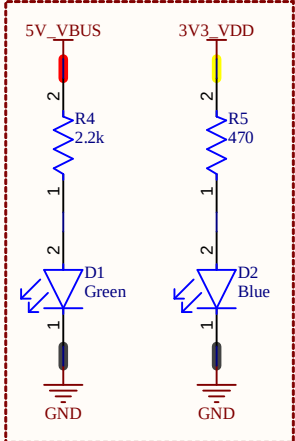
TUSB8044A Decoupling - VDD 3.3V



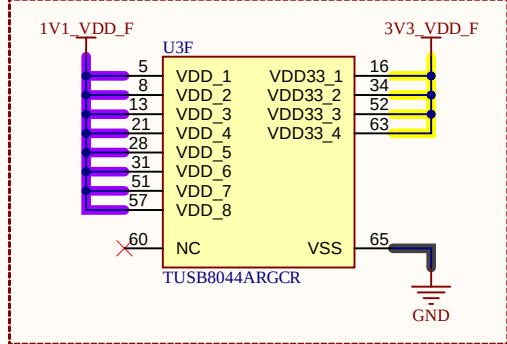
3.3V to 1.1V LDO Regulator



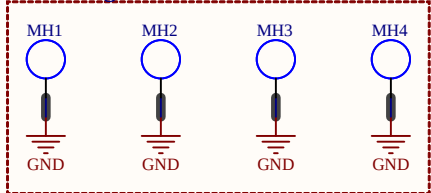
LED Indicators



TUSB8044A Power



Mounting Holes M3



Power-on sequence:
1. VBUS 5V → Buck starts (EN tied to VIN)
2. Buck PG releases → LDO EN goes HIGH (t ≈ 0.5ms)
3. LDO soft-start complete (t ≈ 3.8ms, CSS=2.2nF)
4. LDO PG releases → RC delay charges C_DELAY
5. GRSTz reaches VIH (2V) → TUSB8044A exits reset

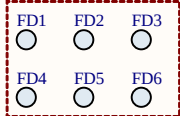
Timing analysis:
t1 = Buck startup to PG: ~0.5ms
t2 = LDO soft-start (VREF×CSS/ISS): 3.3ms
t3 = LDO PG delay (90% of 1.1V): ~0.5ms
t4 = RC delay to VIH: 11.8ms

Total power-on time: ~16ms

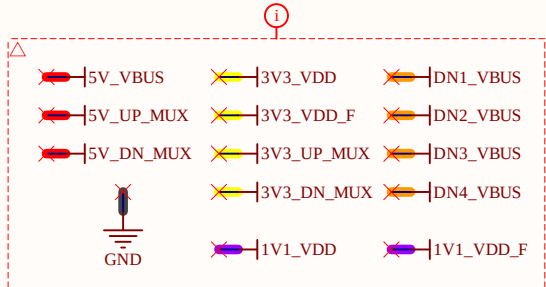
RC delay calculation (GRSTz):
R_ext = 100kΩ (external pull-up on PG)
R_int = 14.5kΩ min (TUSB8044A internal RPU)
R_eq = R_ext // R_int = 12.66kΩ (worst case)
C_DELAY = 1µF
τ = R_eq × C = 12.66ms
t(VIH=2V) = -τ × ln(1 - 2/3.3) = 11.8ms

td2 = 11.8ms > 3ms min (TUSB8044A requirement) ✓

Fiducials



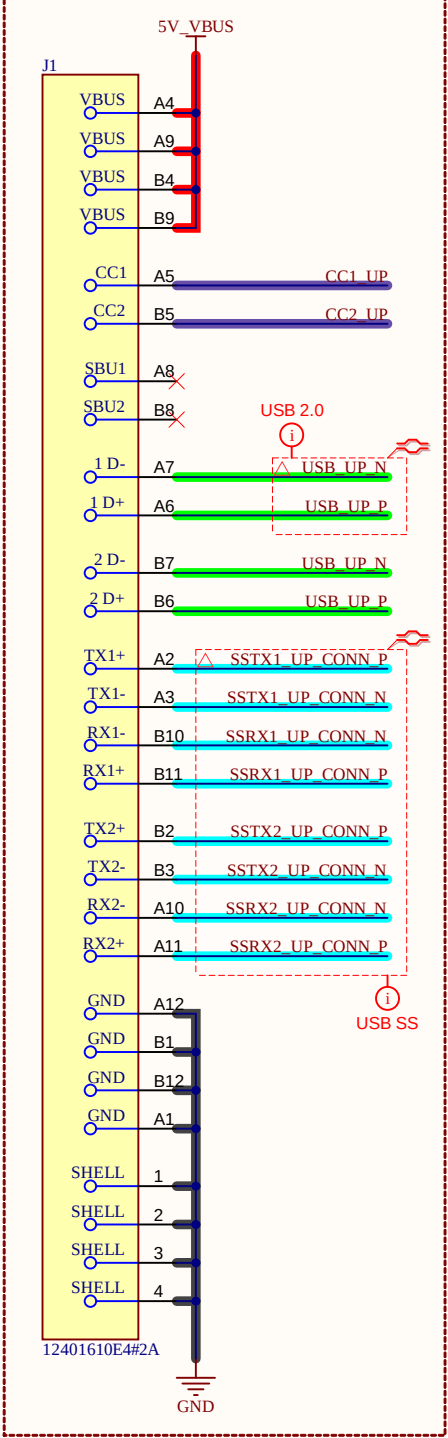
Power Nets



APPROVALS	DATE	PROJECT	Altium™	
ENG: Alberto Marrone		PROJECT REVISION:	DOCUMENT REVISION:	DESIGN ITEM:
DSN: Alberto Marrone		CHK:	Not in version control	Not in version control
REFERENCE DOCUMENTS		TITLE		
BOM:		USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE	CAGE CODE	DWG NO.	REV
FAB DWG:	B			
PCB DWG:	SCALE:	FILE NAME	Power.SchDoc	SHEET 2 OF 7

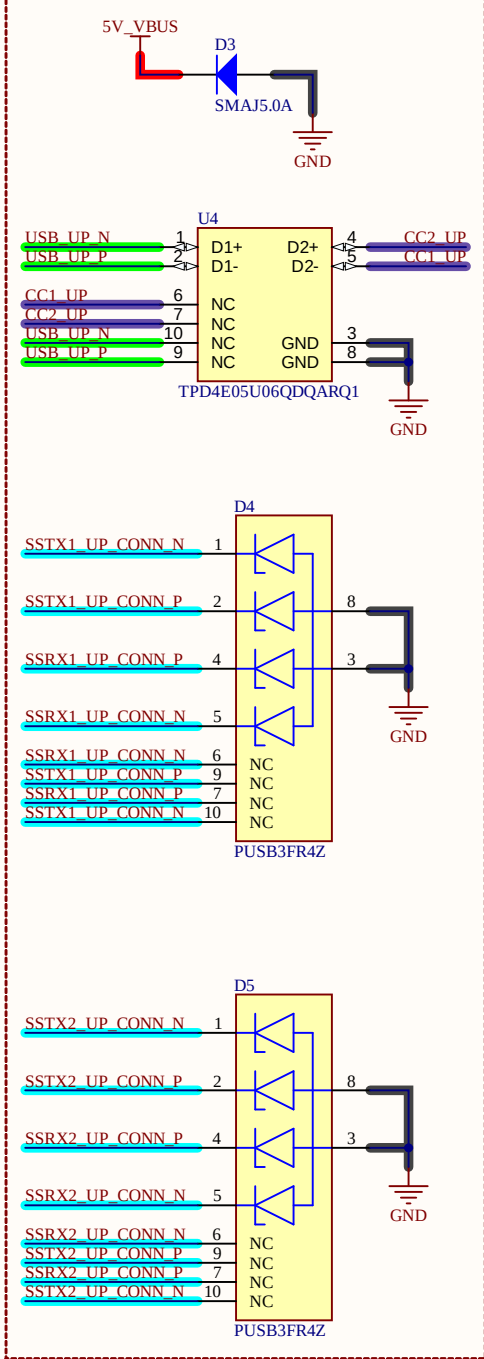
REVISION	DESCRIPTION	DATE	APPROVED

USB-C Receptacle (Upstream)



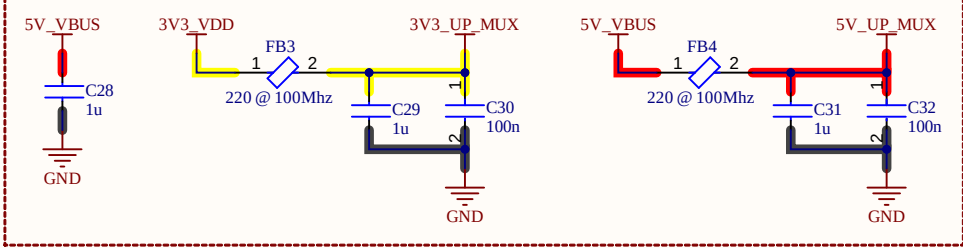
USB-C SBU1/SBU2 left floating
(no Alternate Mode)

ESD Protection

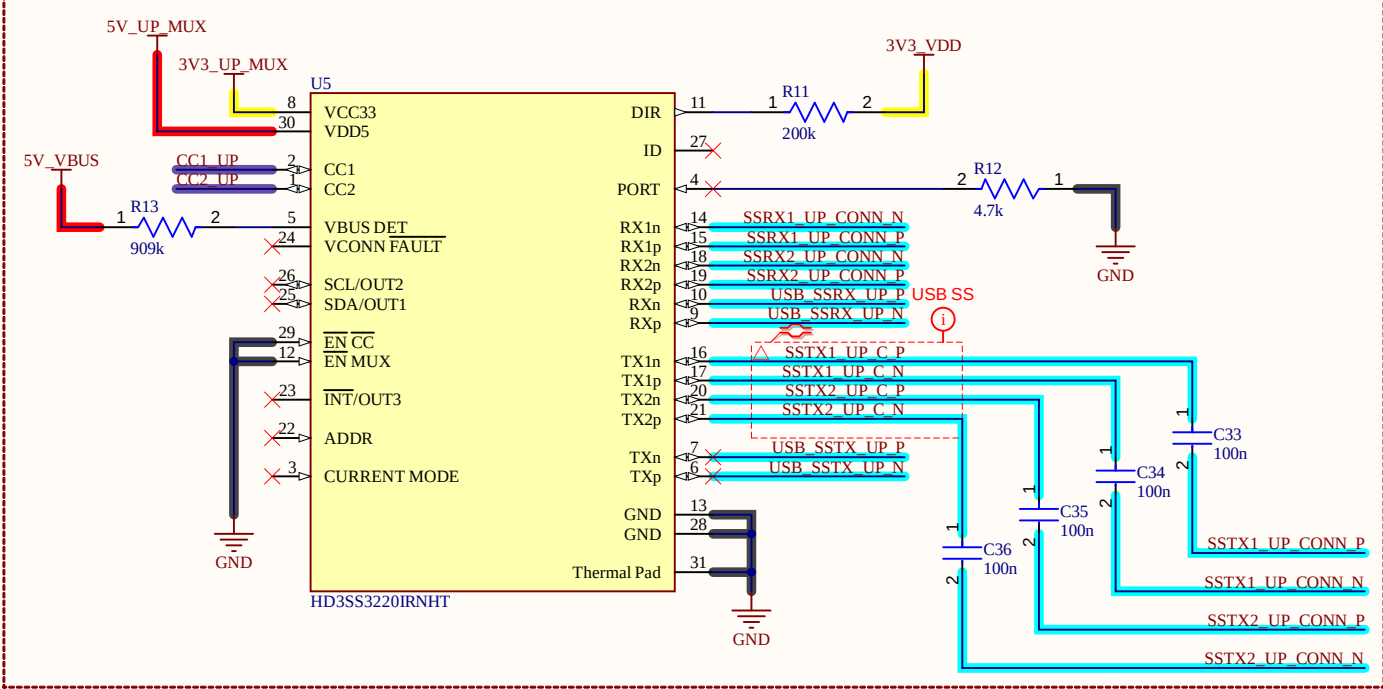


ESD Protection:
- PUSB3FR4Z: 0.29pF, ±15kV, USB3 protection
- TPD4E05U06: 0.5pF, USB2/CC protection
- SMAJ5.0A: VBUS TVS, clamps surges to ground

Power Filtering

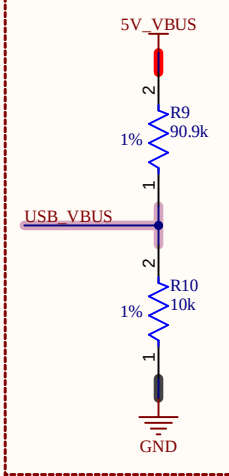


USB-C MUX & CC Controller (UFP Mode)



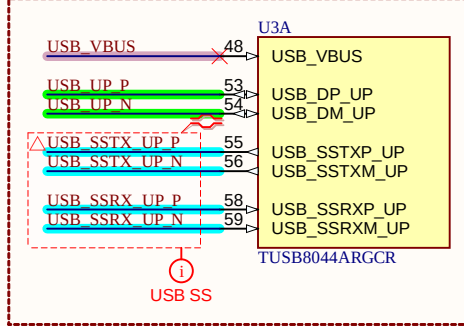
HD3SS3220 Configuration:
- UFP-only mode (PORT pin tied LOW via 4.7kΩ)
- GPIO mode (ADDR floating)
- ENn_CC = ENn_MUX = LOW (always enabled)
- VDD5 stable before VCC33 (datasheet requirement met)

VBUS Detect Divider



VBUS_DET divider:
 $V_{pin} = 5V \times 95k / (909k + 95k) = 0.47V$
Within HD3SS3220 detect range

TUSB8044A Upstream Port



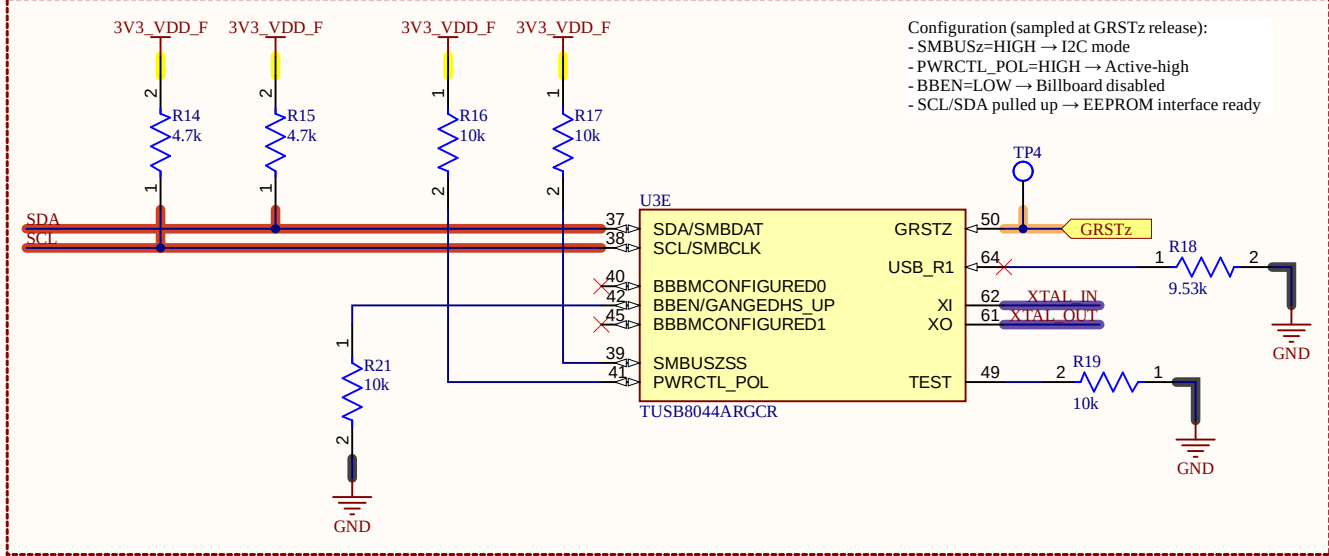
NOTE: SuperSpeed P/N Polarity Swapped for PCB routing optimization.

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DSN: Alberto Marrone		CHK:	Not in version control	Not in version control
REFERENCE DOCUMENTS		TITLE		
BOM:		USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE B	CAGE CODE	DWG NO.	REV
FAB DWG:	SCALE:	FILE NAME	Upstream.SchDoc	SHEET 3 OF 7
PCB DWG:				

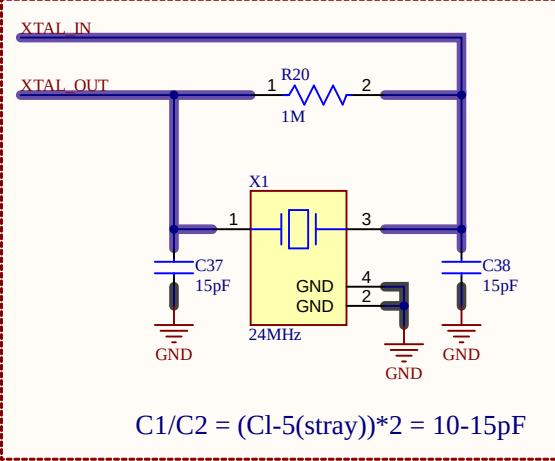
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TUSB8044A Configuration Ports



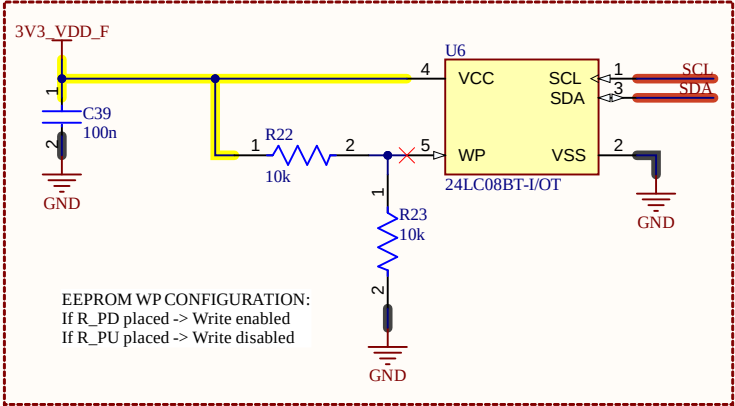
Crystal Oscillator



CRYSTAL PLACEMENT:

- Place within 5mm of TUSB8044A XI/XO pins
- Short, direct traces
- Solid GND plane underneath
- Keep switching signals away
- Guard ring with GND vias recommended

EEPROM

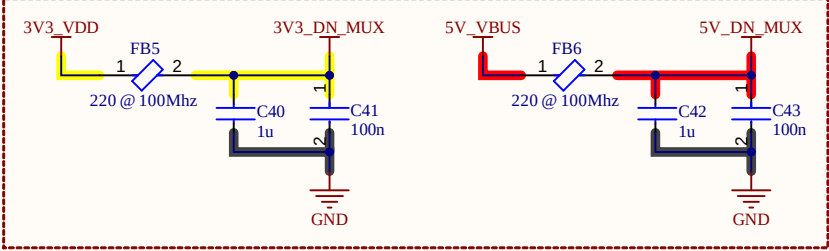


WARNING: Do not populate both resistors simultaneously.

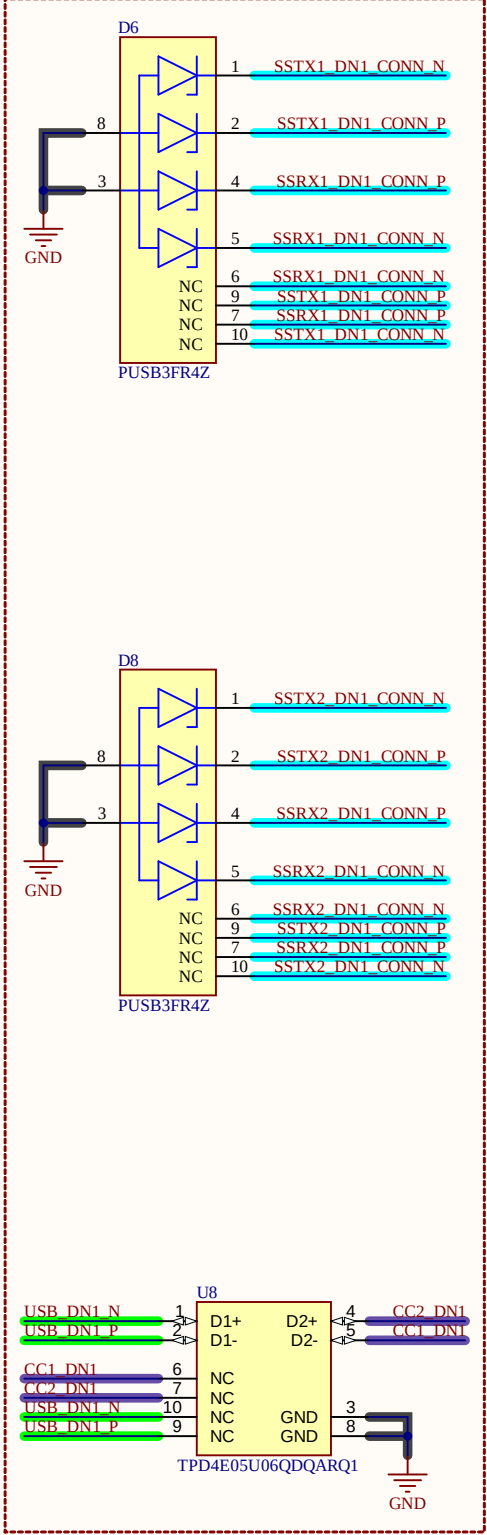
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ENG: Alberto Marrone				
DSN: Alberto Marrone		PROJECT REVISION: Not in version control	DOCUMENT REVISION: Not in version control	DESIGN ITEM: Not in version control
CHK:		TITLE		
REFERENCE DOCUMENTS		USB 3.2 Gen 1 4-Port Hub		
BOM:		SIZE	CAGE CODE	DWG NO.
ASSY DWG:		B		
FAB DWG:				
PCB DWG:		SCALE:	FILE NAME	SHEET 4 OF 7
			Hub Core.SchDoc	

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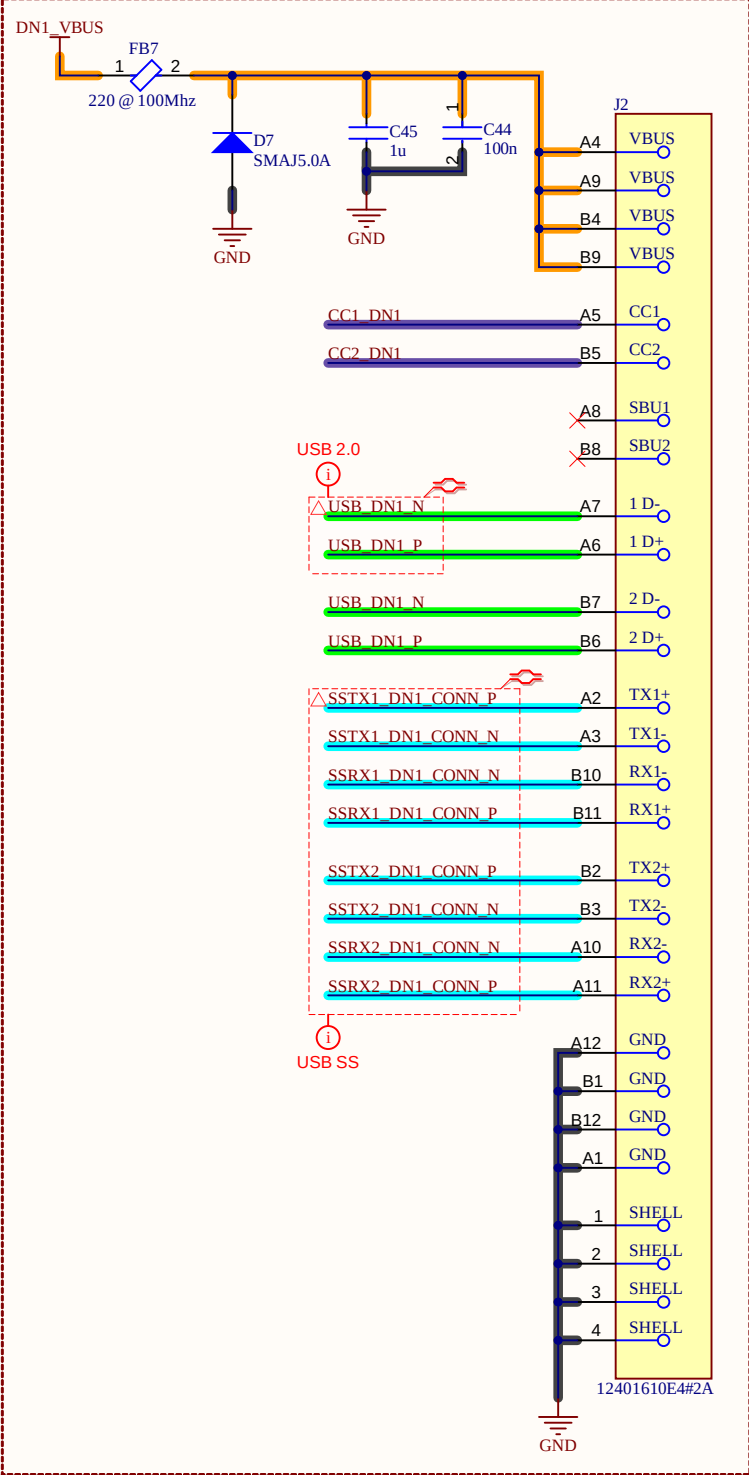
Power Filtering



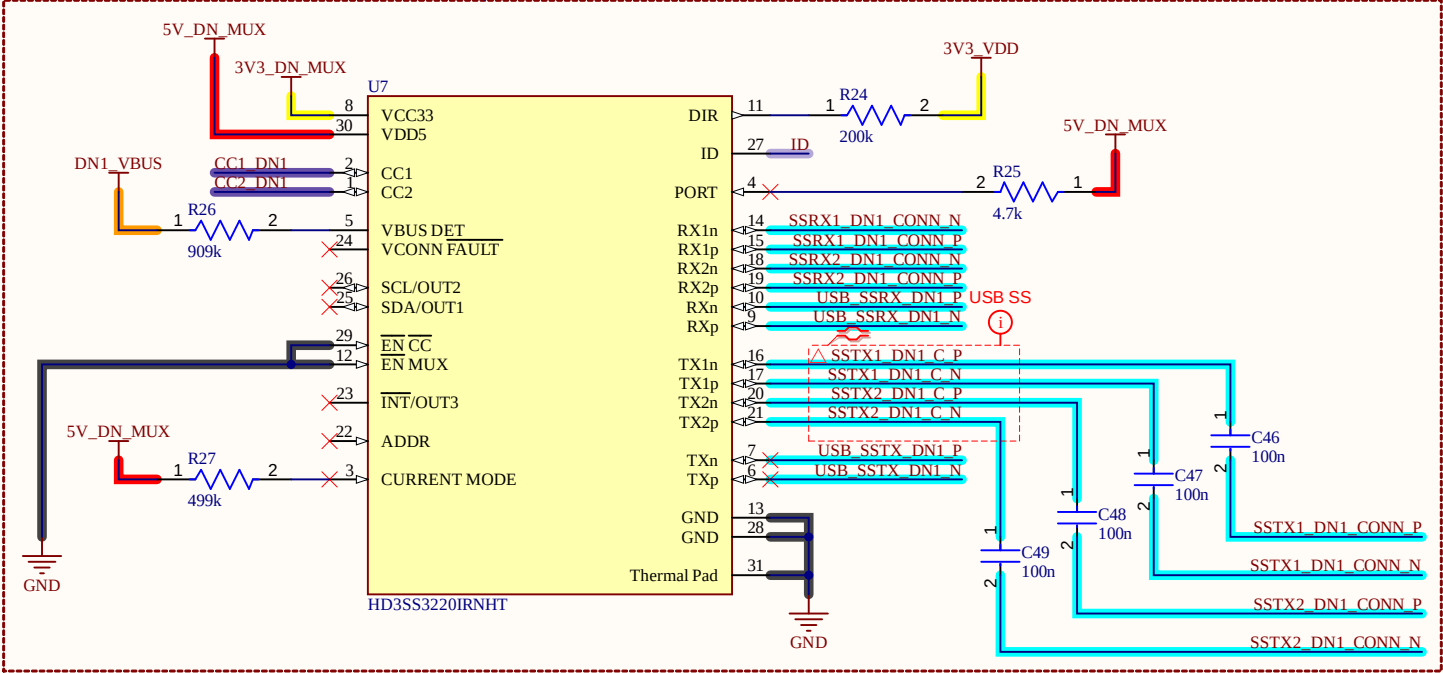
ESD Protection



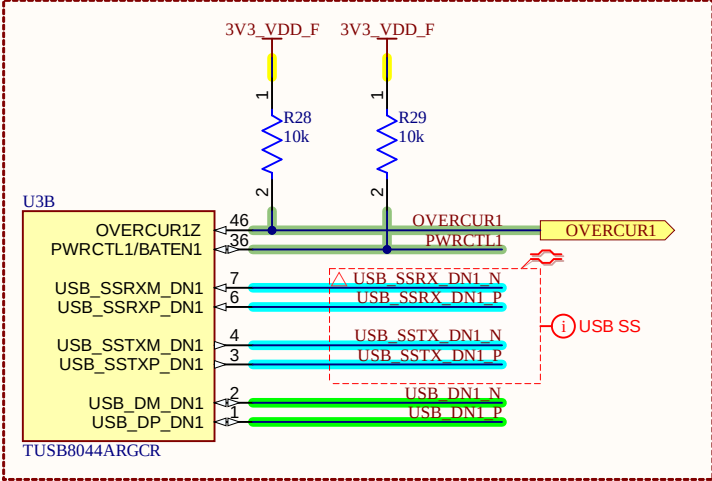
USB-C Receptacle (Downstream Port 1)



USB-C MUX & CC Controller (DFP Mode)

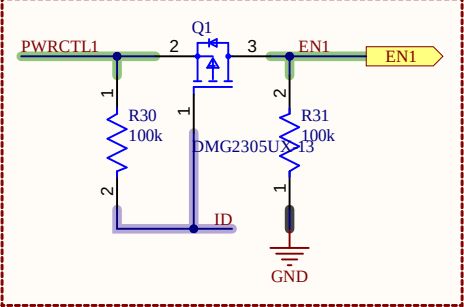


TUSB8044A Downstream Port 1



NOTE: SuperSpeed P/N Polarity Swapped for PCB routing optimization.

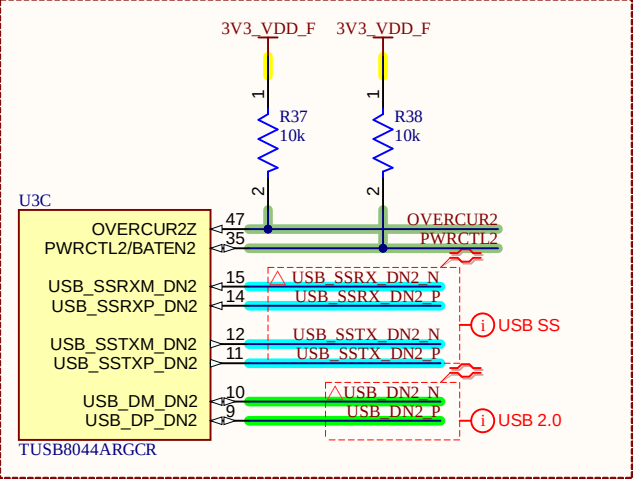
Cold Socket Logic (USB-C Port 1)



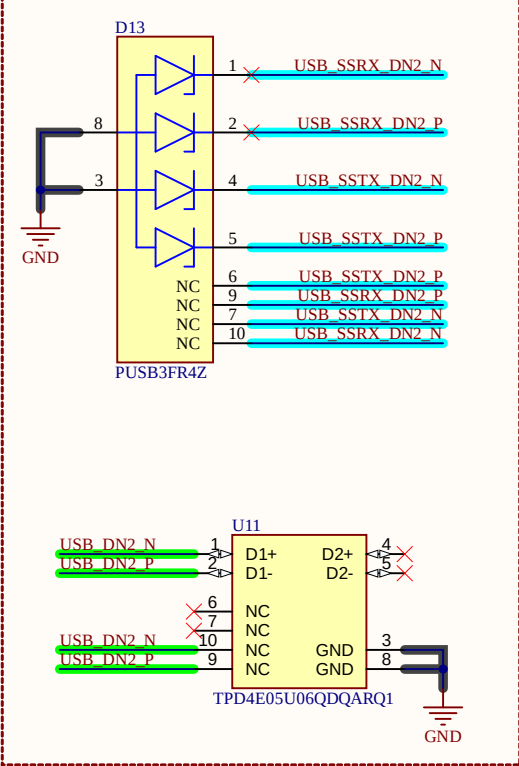
APPROVALS	DATE	PROJECT	Altium	
ENG: Alberto Marrone		PROJECT REVISION:	DOCUMENT REVISION:	DESIGN ITEM:
DSN: Alberto Marrone		CHK:	Not in version control	Not in version control
REFERENCE DOCUMENTS		TITLE		
BOM:		USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE	CAGE CODE	DWG NO.	REV
FAB DWG:	B			
PCB DWG:	SCALE:	FILE NAME	Downstream Port 1.SchDoc	SHEET 5 OF 7

REVISION	DESCRIPTION	DATE	APPROVED

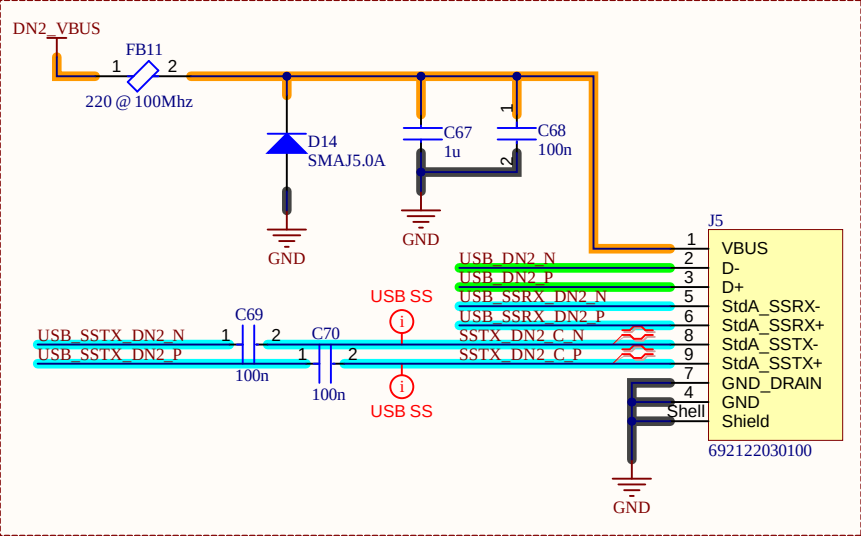
TUSB8044A Downstream Port 2



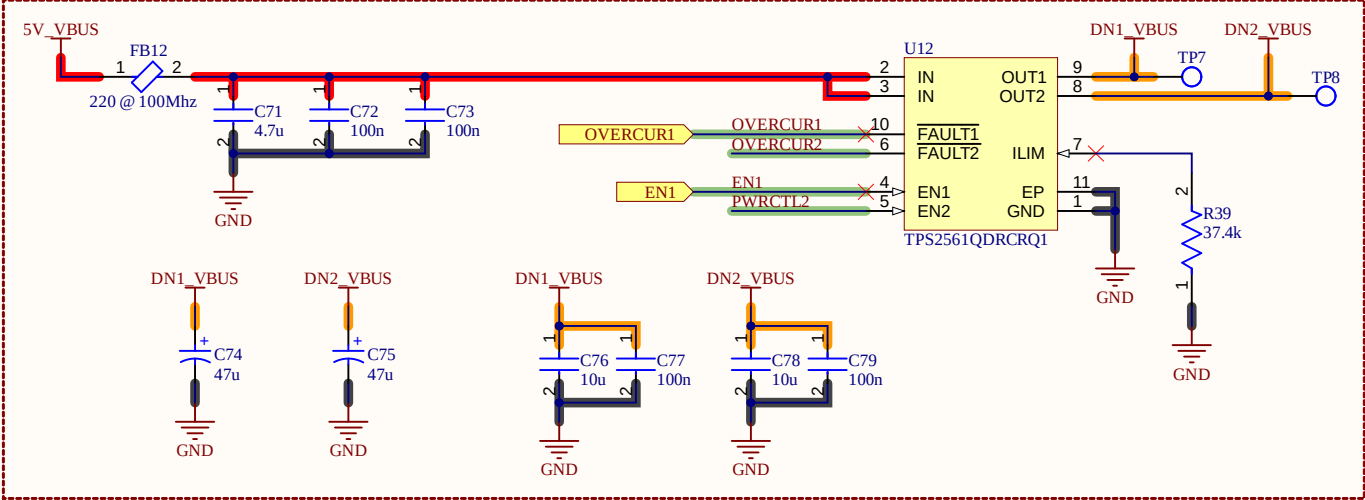
ESD Protection



USB-A Receptacle (Downstream Port 2)

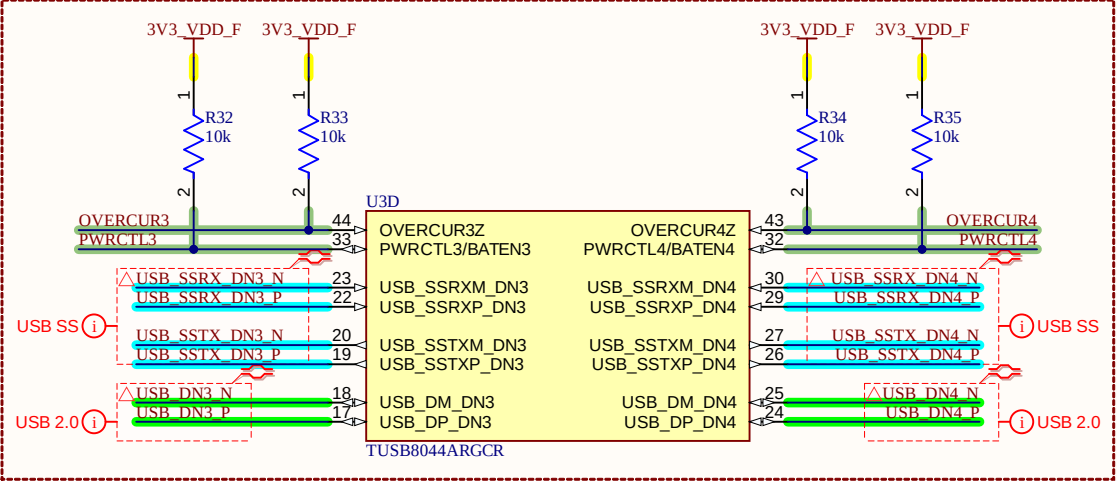


Power Switch & Current Limiting



APPROVALS		DATE		PROJECT 	
ENG: Alberto Marrone					
DSN: Alberto Marrone					
CHK:					
REFERENCE DOCUMENTS				PROJECT REVISION: DOCUMENT REVISION: DESIGN ITEM:	
BOM:				Not in version control Not in version control	
ASSY DWG:				TITLE	
FAB DWG:				USB 3.2 Gen 1 4-Port Hub	
PCB DWG:				SIZE CAGE CODE DWG NO. REV	
SCALE:		FILE NAME		SHEET	
		Downstream Port 2.SchDoc		6 OF 7	

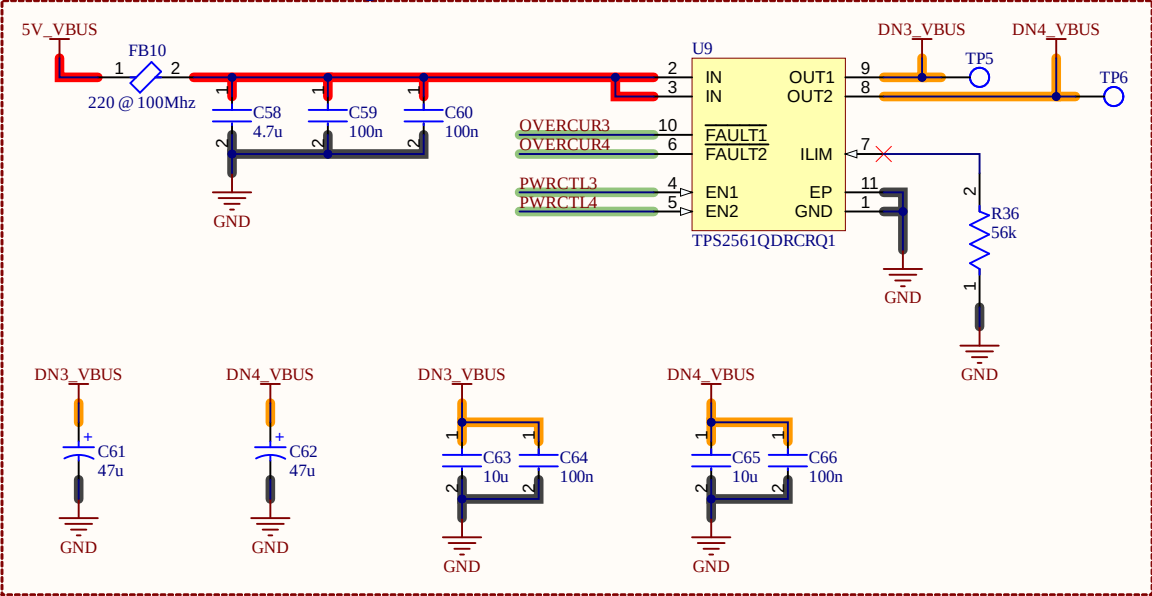
TUSB8044A Downstream Ports 3-4



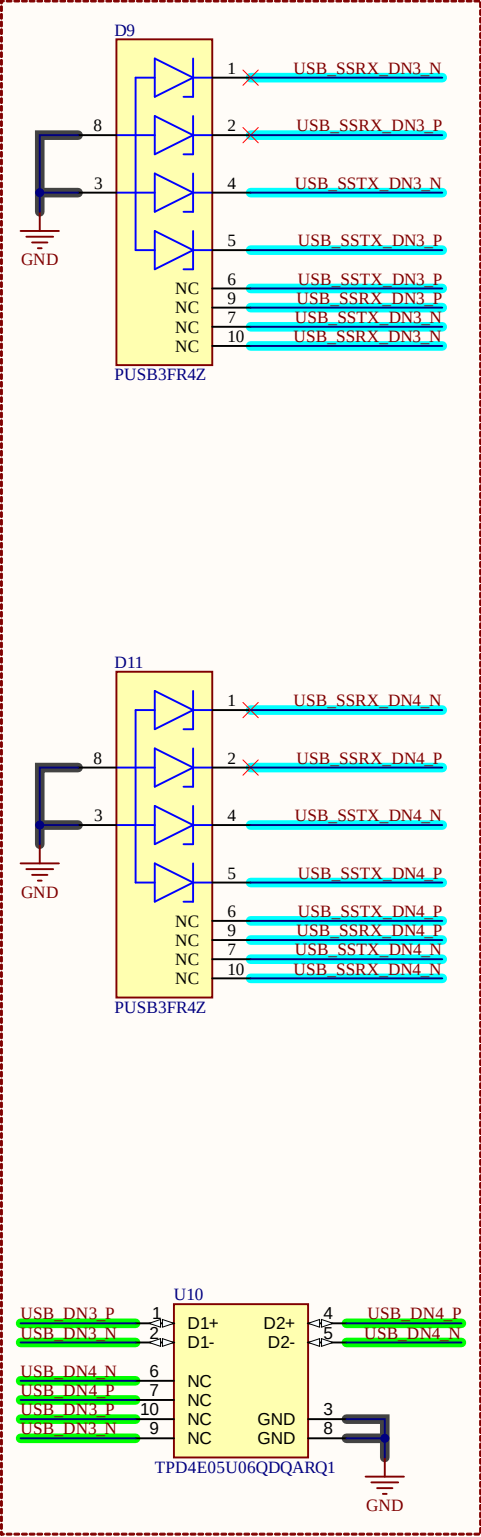
TPS2561 Current Limit Configuration:
 $I_{OS}(typ) = 56000 / R_{ILIM} [k\Omega] (mA)$
 $R_{ILIM} = 37.4k\Omega \rightarrow 1.5A/ch (Ports 1+2)$
 $R_{ILIM} = 56k\Omega \rightarrow 1.0A/ch (Ports 3+4)$

Per-channel limit, NOT shared between channels.
 Overcurrent triggers output disconnect + FAULT LOW.
 Auto-retry after fault condition clears.

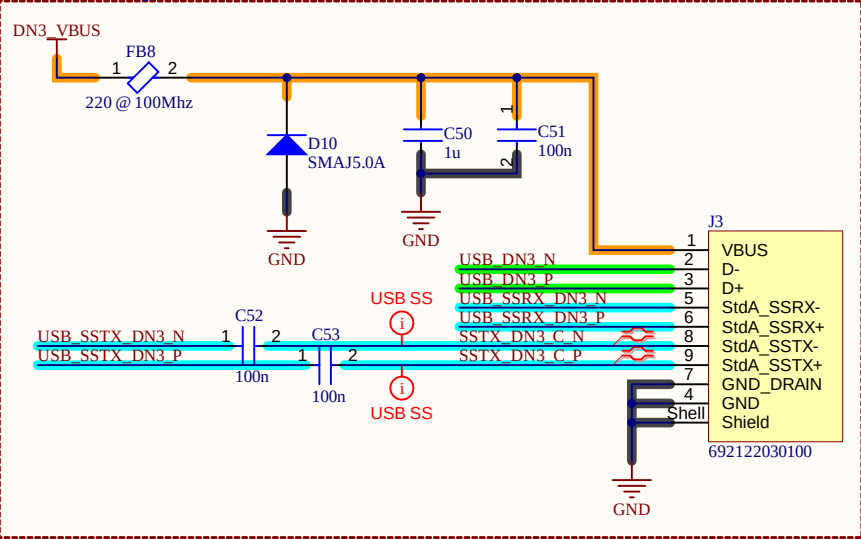
Power Switch & Current Limiting



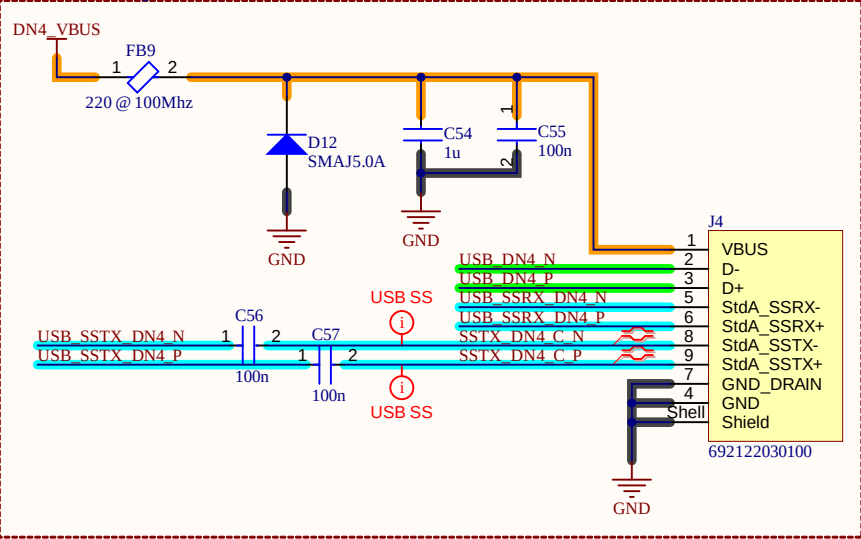
ESD Protection



USB-A Receptacle (Downstream Port 3)



USB-A Receptacle (Downstream Port 4)



APPROVALS		DATE		PROJECT					
ENG: Alberto Marrone									
DSN: Alberto Marrone				PROJECT REVISION:		DOCUMENT REVISION:		DESIGN ITEM:	
CHK:				Not in version control		Not in version control		Not in version control	
REFERENCE DOCUMENTS				TITLE USB 3.2 Gen 1 4-Port Hub					
BOM:									
ASSY DWG:									
FAB DWG:									
PCB DWG:									
SIZE		CAGE CODE		DWG NO.				REV	
B									
SCALE:		FILE NAME		Downstream Ports 3-4.SchDoc				SHEET 7 OF 7	